

PLSQL

Interacting with Oracle Database Server

1. Write a program that displays the Netsal(Sal+comm) of an employee entered at run time from EMP table.(Use NVL Function).

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     v_empno EMP.EMPNO%TYPE := &Emp_No;
  3     v_netsal NUMBER;
  4 BEGIN
  5     SELECT SAL + NVL(COMM,0)
  6     INTO v_netsal
  7     FROM EMP
  8     WHERE EMPNO = v_empno;
  9
 10     DBMS_OUTPUT.PUT_LINE('Net Salary = ' || v_netsal);
 11
 12 EXCEPTION
 13     WHEN NO_DATA_FOUND THEN
 14         DBMS_OUTPUT.PUT_LINE('Employee Not Found');
 15 END;
 16 /
Enter value for emp_no: 7788
old 2:      v_empno EMP.EMPNO%TYPE := &Emp_No;
new 2:      v_empno EMP.EMPNO%TYPE := 7788;
Net Salary = 3000

PL/SQL procedure successfully completed.
```

Writing Control Structures

1. Write a program that prompts for EMPNO, ENAME, SAL. If EMPNO does not exist in EMP table, INSERT the EMPNO, ENAME, SAL into the EMP table; otherwise UPDATE ENAME and SAL for that EMPNO.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     V_EMPNO EMP.EMPNO%TYPE := &EMPNO;
  3     V_ENAME EMP.ENAME%TYPE := '&ENAME';
  4     V_SAL    EMP.SAL%TYPE := &SAL;
  5     CNT NUMBER;
  6 BEGIN
  7     SELECT COUNT(*)
  8     INTO CNT
  9     FROM EMP
 10     WHERE EMPNO = V_EMPNO;
 11
 12     IF CNT = 0 THEN
 13         INSERT INTO EMP(EMPNO, ENAME, SAL)
 14         VALUES(V_EMPNO, V_ENAME, V_SAL);
 15
 16         DBMS_OUTPUT.PUT_LINE('Employee Inserted');
 17     ELSE
 18         UPDATE EMP
 19         SET ENAME = V_ENAME,
 20             SAL = V_SAL
 21         WHERE EMPNO = V_EMPNO;
 22
 23         DBMS_OUTPUT.PUT_LINE('Employee Updated');
 24     END IF;
 25
 26     COMMIT;
 27 END;
 28 /
Enter value for empno: 101
old 2:     V_EMPNO EMP.EMPNO%TYPE := &EMPNO;
new 2:     V_EMPNO EMP.EMPNO%TYPE := 101;
old 3:     V_ENAME EMP.ENAME%TYPE := '&ENAME';
new 3:     V_ENAME EMP.ENAME%TYPE := 'Ward';
Enter value for sal: 10000
old 4:     V_SAL    EMP.SAL%TYPE := &SAL;
new 4:     V_SAL    EMP.SAL%TYPE := 10000;
Employee Inserted
```

2. Write a program that prompts for a number and displays whether the number is ODD or EVEN.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     NUM NUMBER := &NUM;
  3 BEGIN
  4     IF MOD(NUM,2)=0 THEN
  5         DBMS_OUTPUT.PUT_LINE(NUM || ' IS EVEN');
  6     ELSE
  7         DBMS_OUTPUT.PUT_LINE(NUM || ' IS ODD');
  8     END IF;
  9 END;
10 /
Enter value for num: 10
old 2:     NUM NUMBER := &NUM;
new 2:     NUM NUMBER := 10;
10 IS EVEN

PL/SQL procedure successfully completed.

SQL> |
```

3. Write program that prompts for EMPNO and update his salary based on the below specification:

If the employee belongs to DEPTNO 10 then update his salary by 10%.

If the employee belongs to DEPTNO 20 then update his salary by 15%.

If the employee belongs to other DEPTNO then update his salary with SALARY+COMM.

```
SQL> DECLARE
  2     V_EMPNO EMP.EMPNO%TYPE := &EMPNO;
  3     V_DEPTNO EMP.DEPTNO%TYPE;
  4 BEGIN
  5     SELECT DEPTNO
  6     INTO V_DEPTNO
  7     FROM EMP
  8     WHERE EMPNO = V_EMPNO;
  9
 10     IF V_DEPTNO = 10 THEN
 11         UPDATE EMP
 12         SET SAL = SAL + (SAL*0.10)
 13         WHERE EMPNO = V_EMPNO;
 14
 15         DBMS_OUTPUT.PUT_LINE('Salary Updated by 10%');
 16
 17     ELSIF V_DEPTNO = 20 THEN
 18         UPDATE EMP
 19         SET SAL = SAL + (SAL*0.15)
 20         WHERE EMPNO = V_EMPNO;
 21
 22         DBMS_OUTPUT.PUT_LINE('Salary Updated by 15%');
 23
 24     ELSE
 25         UPDATE EMP
 26         SET SAL = SAL + NVL(COMM,0)
 27         WHERE EMPNO = V_EMPNO;
 28
 29         DBMS_OUTPUT.PUT_LINE('Salary Updated with COMM');
 30     END IF;
 31
 32     COMMIT;
 33 END;
 34 /
```

Enter value for empno: 101

old 2: V_EMPNO EMP.EMPNO%TYPE := &EMPNO;

new 2: V_EMPNO EMP.EMPNO%TYPE := 101;

Salary Updated with COMM

PL/SQL procedure successfully completed.

SQL> |

4. In SQL Prompt create a table MYTABLE1 with RESULT as a NUMBER datatype. Write a program that inserts values from 1 to 10 into MYTABLE1. Do not insert 6 and 8.

```
SQL> CREATE TABLE MYTABLE1
 2 (
 3     RESULT NUMBER
 4 );

Table created.

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
 2     I NUMBER := 1;
 3 BEGIN
 4     WHILE I <= 10 LOOP
 5
 6         IF I <> 6 AND I <> 8 THEN
 7             INSERT INTO MYTABLE1
 8                 VALUES(I);
 9         END IF;
10
11         I := I + 1;
12     END LOOP;
13
14     COMMIT;
15
16     DBMS_OUTPUT.PUT_LINE('Records Inserted Successfully');
17 END;
18 /
Records Inserted Successfully

PL/SQL procedure successfully completed.

SQL> SELECT * FROM MYTABLE1;

   RESULT
-----
        1
        2
        3
        4
        5
        7
        9
       10

8 rows selected.
```

Working With Composite Data Types

1. Write a PL/SQL program that displays the entire row of an employee by passing the EMPNO at runtime. (Use %ROWTYPE)

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2     V_EMPNO EMP.EMPNO%TYPE := &EMPNO;
3     EMP_REC EMP%ROWTYPE;
4 BEGIN
5     SELECT *
6     INTO EMP_REC
7     FROM EMP
8     WHERE EMPNO = V_EMPNO;
9
10    DBMS_OUTPUT.PUT_LINE('Employee Number : ' || EMP_REC.EMPNO);
11    DBMS_OUTPUT.PUT_LINE('Employee Name : ' || EMP_REC.ENAME);
12    DBMS_OUTPUT.PUT_LINE('Job : ' || EMP_REC.JOB);
13    DBMS_OUTPUT.PUT_LINE('Manager : ' || EMP_REC.MGR);
14    DBMS_OUTPUT.PUT_LINE('Hire Date : ' || EMP_REC.HIREDATE);
15    DBMS_OUTPUT.PUT_LINE('Salary : ' || EMP_REC.SAL);
16    DBMS_OUTPUT.PUT_LINE('Commission : ' || NVL(EMP_REC.COMM,0));
17    DBMS_OUTPUT.PUT_LINE('Department No : ' || EMP_REC.DEPTNO);
18
19 EXCEPTION
20     WHEN NO_DATA_FOUND THEN
21         DBMS_OUTPUT.PUT_LINE('Employee Not Found.');
```

22 END;

23 /

Enter value for empno: 101

old 2: V_EMPNO EMP.EMPNO%TYPE := &EMPNO;

new 2: V_EMPNO EMP.EMPNO%TYPE := 101;

Employee Number : 101

Employee Name : Ward

Job :

Manager :

Hire Date :

Salary : 10000

Commission : 0

Department No :

PL/SQL procedure successfully completed.

Explicit Cursors

1. Write an implicit cursor program that prompts for a JOB and deletes the employees working in that job. Store the number of records deleted in a session variable.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     V_JOB EMP.JOB%TYPE := UPPER('&JOB');
  3     V_COUNT NUMBER;
  4 BEGIN
  5     DELETE FROM EMP
  6     WHERE JOB = V_JOB;
  7
  8     V_COUNT := SQL%ROWCOUNT;
  9
 10     DBMS_OUTPUT.PUT_LINE(V_COUNT || ' Record(s) Deleted. ');
 11
 12     COMMIT;
 13 END;
 14 /
Enter value for job: analyst
old 2:     V_JOB EMP.JOB%TYPE := UPPER('&JOB');
new 2:     V_JOB EMP.JOB%TYPE := UPPER('analyst');
2 Record(s) Deleted.

PL/SQL procedure successfully completed.

SQL> |
```

2. Rollback the previous DELETE statement.

```
SQL> BEGIN
  2     ROLLBACK;
  3     DBMS_OUTPUT.PUT_LINE('Rollback Successful');
  4 END;
  5 /
Rollback Successful

PL/SQL procedure successfully completed.

SQL> |
```

3. Write a program that displays the Top N salaries using a Simple Explicit Cursor.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     V_N NUMBER := &N;
  3     V_COUNT NUMBER := 0;
  4
  5     CURSOR C1 IS
  6     SELECT EMPNO, ENAME, SAL
  7     FROM EMP
  8     ORDER BY SAL DESC;
  9 BEGIN
 10     FOR REC IN C1 LOOP
 11         EXIT WHEN V_COUNT = V_N;
 12
 13         DBMS_OUTPUT.PUT_LINE(
 14             REC.EMPNO || ' ' ||
 15             REC.ENAME || ' ' ||
 16             REC.SAL);
 17
 18         V_COUNT := V_COUNT + 1;
 19     END LOOP;
 20 END;
 21 /
Enter value for n: 5
old  2:     V_N NUMBER := &N;
new  2:     V_N NUMBER := 5;
101 Ward 10000
7839 KING 5000
7566 JONES 2975
7698 BLAKE 2850
7782 CLARK 2450

PL/SQL procedure successfully completed.

SQL> |
```


4. Create a Simple Explicit Cursor that displays all the employees who are earning more than the average salary of the EMP table.

```
SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2     V_AVG EMP.SAL%TYPE;
  3     V_EMPNO EMP.EMPNO%TYPE;
  4     V_ENAME EMP.ENAME%TYPE;
  5     V_SAL EMP.SAL%TYPE;
  6
  7     CURSOR C1 IS
  8         SELECT EMPNO, ENAME, SAL
  9         FROM EMP
 10         WHERE SAL > V_AVG;
 11
 12 BEGIN
 13     SELECT AVG(SAL)
 14     INTO V_AVG
 15     FROM EMP;
 16
 17     DBMS_OUTPUT.PUT_LINE('Average Salary = ' || ROUND(V_AVG,2));
 18
 19     OPEN C1;
 20
 21     LOOP
 22         FETCH C1 INTO V_EMPNO, V_ENAME, V_SAL;
 23         EXIT WHEN C1%NOTFOUND;
 24
 25         DBMS_OUTPUT.PUT_LINE(
 26             V_EMPNO || ' ' || V_ENAME || ' ' || V_SAL
 27             V_ENAME || ' ' || V_SAL
 28             V_SAL
 29         );
 30     END LOOP;
 31
 32     CLOSE C1;
 33 END;
 34 /
Average Salary = 2048.33
101 KING 5000
102 SCOTLER 3000
105 BLAKE 2850
106 JONES 2975
112 CLARKEN 2450
114 SARALA 2100
115 RAMESH 2600

PL/SQL procedure successfully completed.
```

5. Create a Simple Explicit cursor that displays the employees who earn more than 2000 and have joined the company after 15-Jun-1981. Print the output : <> earns <> and joined the organization on <>.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2   CURSOR C1 IS
3     SELECT ENAME, SAL, HIREDATE
4     FROM EMP
5     WHERE SAL > 2000
6           AND HIREDATE > TO_DATE('15-JUN-1981','DD-MON-YYYY');
7
8   V_ENAME EMP.ENAME%TYPE;
9   V_SAL EMP.SAL%TYPE;
10  V_HIREDATE EMP.HIREDATE%TYPE;
11
12 BEGIN
13   OPEN C1;
14
15   LOOP
16     FETCH C1 INTO V_ENAME, V_SAL, V_HIREDATE;
17     EXIT WHEN C1%NOTFOUND;
18
19     DBMS_OUTPUT.PUT_LINE(
20       V_ENAME || ' earns ' || V_SAL ||
21       ' and joined the organization on ' ||
22       TO_CHAR(V_HIREDATE,'DD-MON-YYYY')
23     );
24   END LOOP;
25
26   CLOSE C1;
27 END;
28 /
```

KING earns 5000 and joined the organization on 17-NOV-1981

PL/SQL procedure successfully completed.

SQL> |

6.Create a Simple Explicit Cursor that displays the Hire Date in the format DD-MM-RRRR and Day. Sort the records on Day, starting from Saturday.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2   CURSOR C1 IS
3   SELECT ENAME,
4          TO_CHAR(HIREDATE, 'DD-MM-RRRR') HDATE,
5          TO_CHAR(HIREDATE, 'DAY') DAYNAME
6   FROM EMP
7   ORDER BY TO_CHAR(HIREDATE, 'D');
8
9   V_ENAME EMP.ENAME%TYPE;
10  V_HDATE VARCHAR2(20);
11  V_DAY VARCHAR2(20);
12
13 BEGIN
14   OPEN C1;
15
16   LOOP
17     FETCH C1 INTO V_ENAME, V_HDATE, V_DAY;
18     EXIT WHEN C1%NOTFOUND;
19
20     DBMS_OUTPUT.PUT_LINE(
21       RPAD(V_ENAME,10) || ' ' ||
22       V_HDATE || ' ' ||
23       RTRIM(V_DAY)
24     );
25   END LOOP;
26
27   CLOSE C1;
28 END;
29 /
WARD      22-02-1981 SUNDAY
MARTIN    28-09-1981 MONDAY
CLARK     09-06-1981 TUESDAY
TURNER    08-09-1981 TUESDAY
KING      17-11-1981 TUESDAY
ADAMS     12-01-1983 WEDNESDAY
SMITH     17-12-1980 WEDNESDAY
JAMES     03-12-1981 THURSDAY
JONES     02-04-1981 THURSDAY
BLAKE     01-05-1981 FRIDAY
ALLEN     20-02-1981 FRIDAY
MILLER    23-01-1982 SATURDAY
Ward
```

PL/SQL procedure successfully completed.

7. Create a Simple Explicit Cursor that displays the employee names and commission amounts. If an employee does not earn commission, display "No Commission". Label the commission column as COMM.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2   CURSOR C1 IS
  3   SELECT ENAME,
  4          NVL(TO_CHAR(COMM), 'No Commission') COMM
  5   FROM EMP;
  6
  7   V_ENAME EMP.ENAME%TYPE;
  8   V_COMM VARCHAR2(30);
  9
 10 BEGIN
 11   OPEN C1;
 12
 13   LOOP
 14     FETCH C1 INTO V_ENAME, V_COMM;
 15     EXIT WHEN C1%NOTFOUND;
 16
 17     DBMS_OUTPUT.PUT_LINE(
 18       RPAD(V_ENAME,10) || ' COMM : ' || V_COMM
 19     );
 20   END LOOP;
 21
 22   CLOSE C1;
 23 END;
 24 /
SMITH      COMM : No Commission
ALLEN      COMM : 300
WARD       COMM : 500
JONES      COMM : No Commission
MARTIN     COMM : 1400
BLAKE      COMM : No Commission
CLARK      COMM : No Commission
KING       COMM : No Commission
TURNER     COMM : 0
ADAMS      COMM : No Commission
JAMES      COMM : No Commission
MILLER     COMM : No Commission
Ward       COMM : No Commission
```

PL/SQL procedure successfully completed.

8. Create a parameter cursor that takes DEPTNO as a parameter and update the star column with '*'. Every '*' represent a 1000. For example : If the employees salary is 5000 then update star column with '*****'. Use Parameter Cursor ,CURSOR WITH FOR LOOP and WHERE CURRENT OF OPTION.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     V_DEPTNO NUMBER := &DEPTNO;
  3
  4     CURSOR C1(P_DNO NUMBER) IS
  5         SELECT SAL, STAR
  6         FROM EMP
  7         WHERE DEPTNO = P_DNO
  8         FOR UPDATE OF STAR;
  9
 10 BEGIN
 11     FOR REC IN C1(V_DEPTNO) LOOP
 12
 13         UPDATE EMP
 14         SET STAR = RPAD('*', TRUNC(REC.SAL/1000), '*')
 15         WHERE CURRENT OF C1;
 16
 17     END LOOP;
 18
 19     COMMIT;
 20
 21     DBMS_OUTPUT.PUT_LINE('STAR column updated successfully.');
```

22 END;

23 /

Enter value for deptno: 10

old 2: V_DEPTNO NUMBER := &DEPTNO;

new 2: V_DEPTNO NUMBER := 10;

STAR column updated successfully.

PL/SQL procedure successfully completed.

9. Write a Parameter Cursor program that promotes CLERK who earn more than 1000 to SR CLERK and increase the salary by 10%. Pass CLERK as a parameter to the Cursor. Use Parameter Cursor, CURSOR WITH FOR LOOP and CURSOR WITH UPDATE CLAUSE.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2
3     CURSOR C1(P_JOB VARCHAR2) IS
4         SELECT JOB, SAL
5         FROM EMP
6         WHERE JOB=P_JOB
7         AND SAL>1000
8         FOR UPDATE OF JOB, SAL;
9
10 BEGIN
11
12     FOR REC IN C1('CLERK') LOOP
13
14         UPDATE EMP
15         SET JOB='SR CLERK',
16             SAL=SAL*1.10
17         WHERE CURRENT OF C1;
18
19     END LOOP;
20
21     COMMIT;
22
23     DBMS_OUTPUT.PUT_LINE('Employees Promoted Successfully.');
```

24

25 END;

26 /

Employees Promoted Successfully.

PL/SQL procedure successfully completed.

SQL> |

10. Display entire record from the Dept table by passing a DEPTNO. Increase the Salary of employees working in deptno 10 by 15%, Deptno 20 by 15% and others by 5%. Also display the corresponding employees working in that Dept. Use a parameter Cursor, For Loop Cursor and Cursor with Update clause.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2
  3     V_DEPTNO NUMBER := &DEPTNO;
  4
  5     V_DNAME DEPT.DNAME%TYPE;
  6     V_LOC DEPT.LOC%TYPE;
  7
  8     CURSOR C1(P_DNO NUMBER) IS
  9         SELECT EMPNO, ENAME, SAL, DEPTNO
 10         FROM EMP
 11         WHERE DEPTNO=P_DNO
 12         FOR UPDATE OF SAL;
 13
 14 BEGIN
 15
 16     SELECT DNAME, LOC
 17     INTO V_DNAME, V_LOC
 18     FROM DEPT
 19     WHERE DEPTNO=V_DEPTNO;
 20
 21     DBMS_OUTPUT.PUT_LINE('Department No : ' || V_DEPTNO);
 22     DBMS_OUTPUT.PUT_LINE('Department   : ' || V_DNAME);
 23     DBMS_OUTPUT.PUT_LINE('Location     : ' || V_LOC);
 24
 25     DBMS_OUTPUT.PUT_LINE('-----');
 26
 27     FOR REC IN C1(V_DEPTNO) LOOP
 28
 29         IF REC.DEPTNO=10 THEN
 30
 31             UPDATE EMP
 32             SET SAL=SAL*1.15
 33             WHERE CURRENT OF C1;
 34
 35         ELSIF REC.DEPTNO=20 THEN
 36
 37             UPDATE EMP
 38             SET SAL=SAL*1.15
 39             WHERE CURRENT OF C1;
 40
 41         ELSE
 42
```

```
43          UPDATE EMP
44          SET SAL=SAL*1.05
45          WHERE CURRENT OF C1;
46
47      END IF;
48
49      DBMS_OUTPUT.PUT_LINE(
50          REC.EMPNO||' '||
51          REC.ENAME||' Salary Updated');
52
53  END LOOP;
54
55  COMMIT;
56
57  END;
58  /
```

Enter value for deptno: 20

old 3: V_DEPTNO NUMBER := &DEPTNO;

new 3: V_DEPTNO NUMBER := 20;

Department No : 20

Department : RESEARCH

Location : DALLAS

113 ADAMSSMITH Salary Updated

102 SCOTLER Salary Updated

104 SMITHKO Salary Updated

106 JONES Salary Updated

115 RAMESH Salary Updated

PL/SQL procedure successfully completed.

11. Write a program that will prompt the user for a CHOICE. IF CHOICE is 1 then display all the employees who earn more than 2000 from EMP table else display all the Employees who earn less than 2000. Use REF CURSORS.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2     CHOICE NUMBER := &CHOICE;
3
4     TYPE EMP_CUR IS REF CURSOR;
5     C1 EMP_CUR;
6
7     V_EMPNO EMP.EMPNO%TYPE;
8     V_ENAME EMP.ENAME%TYPE;
9     V_SAL EMP.SAL%TYPE;
10
11 BEGIN
12
13     IF CHOICE = 1 THEN
14         OPEN C1 FOR
15             SELECT EMPNO, ENAME, SAL
16             FROM EMP
17             WHERE SAL > 2000;
18     ELSE
19         OPEN C1 FOR
20             SELECT EMPNO, ENAME, SAL
21             FROM EMP
22             WHERE SAL < 2000;
23     END IF;
24
25     LOOP
26         FETCH C1 INTO V_EMPNO, V_ENAME, V_SAL;
27         EXIT WHEN C1%NOTFOUND;
28
29         DBMS_OUTPUT.PUT_LINE(
30             'Emp No : ' || V_EMPNO ||
31             ' Name : ' || V_ENAME ||
32             ' Salary : ' || V_SAL
33         );
34     END LOOP;
35
36     CLOSE C1;
37
38 END;
39 /
Enter value for choice: 10
old 2:  CHOICE NUMBER := &CHOICE;
new 2:  CHOICE NUMBER := 10;
Emp No : 7369 Name : SMITH Salary : 800
Emp No : 7499 Name : ALLEN Salary : 1600
Emp No : 7521 Name : WARD Salary : 1250
Emp No : 7654 Name : MARTIN Salary : 1250
Emp No : 7844 Name : TURNER Salary : 1500
Emp No : 7876 Name : ADAMS Salary : 1210
Emp No : 7900 Name : JAMES Salary : 950
Emp No : 7934 Name : MILLER Salary : 1644.5

PL/SQL procedure successfully completed.

SQL> |
```